



Terascale Clusters and Data Grids: A Look at the Future

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The HPC Conundrum

- **The conundrum**

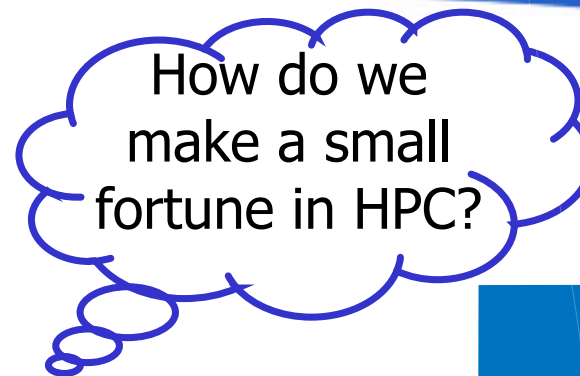
- large NRE costs
 - particularly software
- modest size markets
 - TMC, KSR, Cray, SGI, ...
- NCSA's previous vendors
 - almost all RIP

- **Implications**

- 1980s-1990s market shakeout
- limited base to amortize software costs

- **Rescue (Ben Franklin)**

- open source
- commodity clusters



It's The Software, Duh

- **We have this fascination with hardware**
 - it's tangible and obvious
 - it's necessary but not sufficient
 - like a car without a road
 - look at our experiences
 - CDC 6600, Cray 1, CM-5, KSR, ...
- **Software is always overlooked**
 - both application and infrastructure
 - application embodies the science
 - tools enable/hinder productivity
 - it's not cheap!
 - think about the cost of a single Internet startup **L**
 - you must keep supporting it



Why Linux Clusters?

- “Every

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- Thriving

- com
- a
- ven



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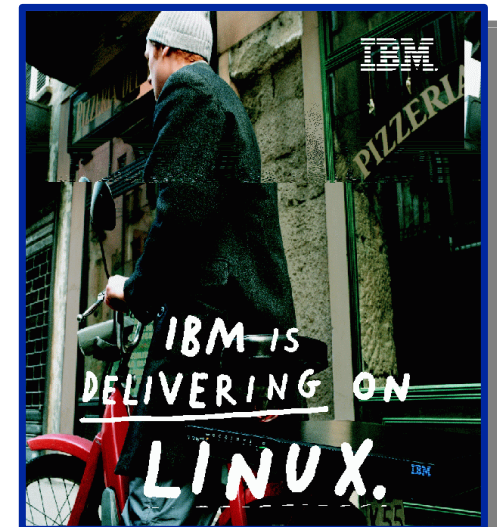
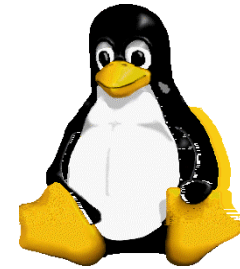
vehicles

roups

community

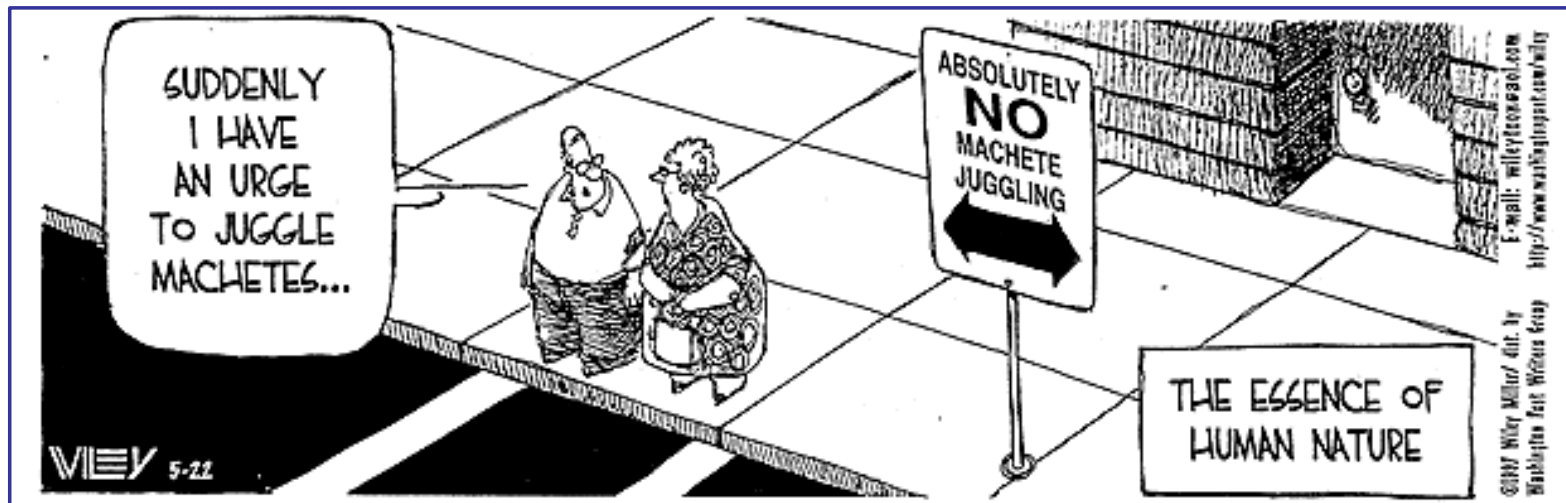
nt

and tools



Similar ad campaigns
for other vendors

On The Other Hand ...



... we could just be crazy! J

But We Don't

Terascale Clusters: First Steps

- **1 TF IA-32 Pentium III cluster (Platinum)**

- 512 1 GHz dual processor nodes
- Myrinet 2000 interconnect
- 5 TB of RAID storage
- #30 on Top500 list



- **1 TF IA-64 Itanium cluster (Titan)**

- 164 800 MHz dual processor nodes
- Myrinet 2000 interconnect
- shared disk storage



NCSA Platinum Cluster Timeline

**2/23 First four racks
of IBM hardware arrive**

4/4 NFS problem resolved for CMS
4/5 Myrinet static mapping in place
4/7 CMS runs successfully
4/11 400 processor HPL runs completing
4/12 Myricom engineering assistance

**6/1 Friendly user
period begins**

Jan 2001

Feb 2001

Mar 2001

Apr 2001

May 2001

June 2001

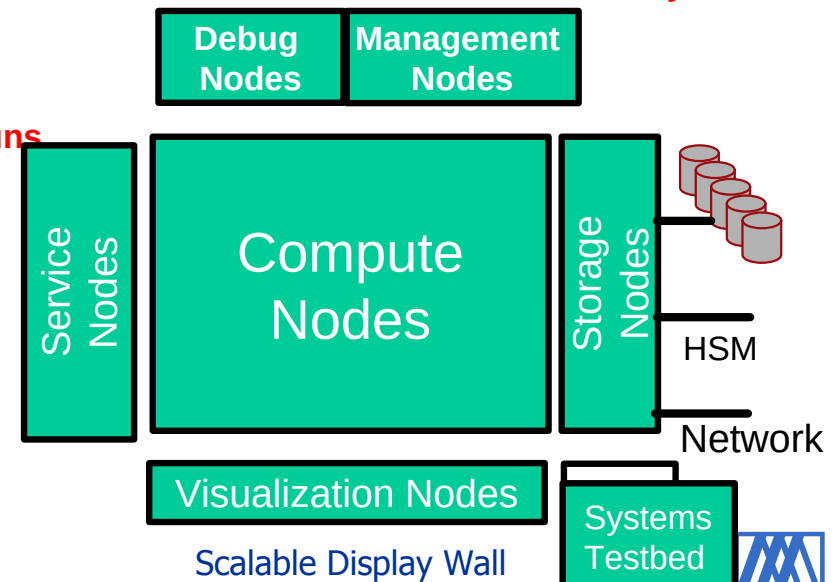
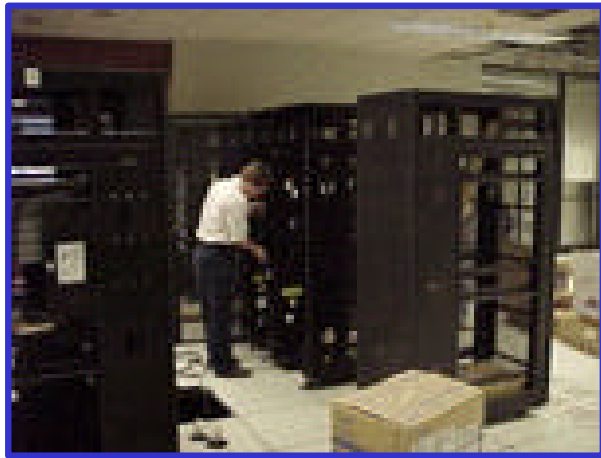
July 2001

Order placed
with IBM
512 compute
node cluster

3/1 Head nodes operational
3/10 First 126 processor Myrinet test jobs
3/13 Final IBM hardware shipment
**3/22 First application for compute
nodes (CMS/Koranda/Litvin)**
3/26 Initial Globus installation
3/26 Final Myrinet hardware arrives
3/26 First 512 processor MILC and NAMD runs

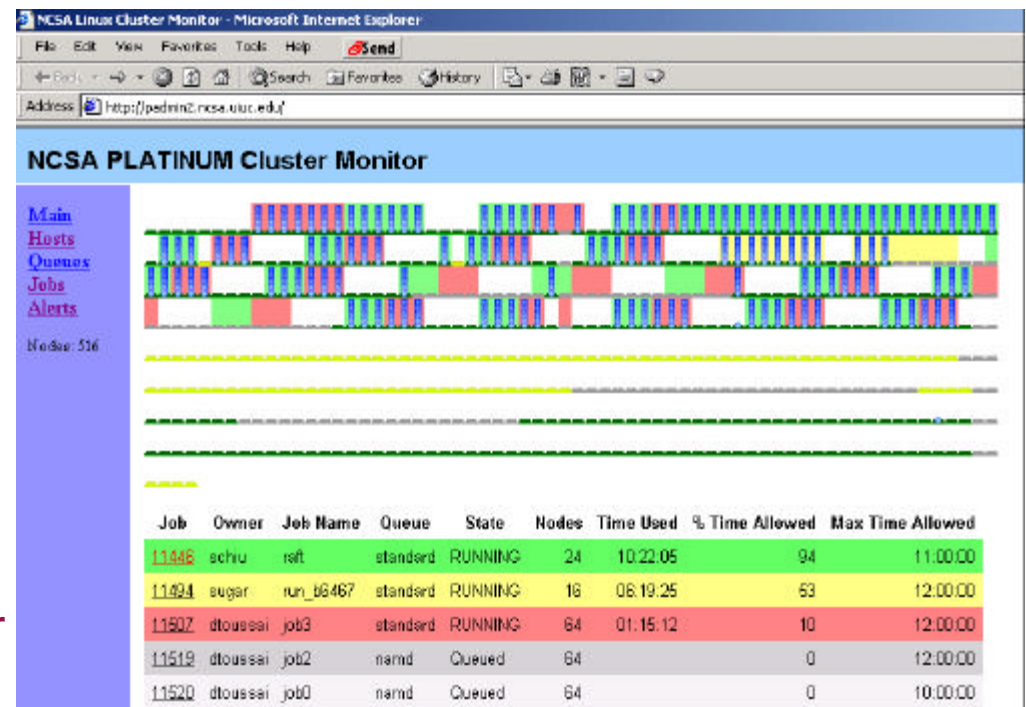
5/8 1000p MP Linpack runs
5/11 1008 processor Top500 run @ 594GF
5/14 2.4 Kernel testing
5/28 RedHat 7.1 testing

**Production service
since July**



Platinum Software Configuration

- **Linux**
 - RedHat 6.2 and Linux 2.2.19 SMP Kernel
- **Open PBS**
 - resource management and job control
- **Maui Scheduler**
 - advanced scheduling
- **Argonne MPICH**
 - parallel programming API
- **NCSA VMI**
 - communication middleware
 - MPICH and Myrinet
- **Myricom GM**
 - Myrinet communication layer
- **NCSA cluster monitor**



The Biology Revolution

DNA
sequence

Protein sequence
and regulation

Protein
structure

Protein/enzyme
function



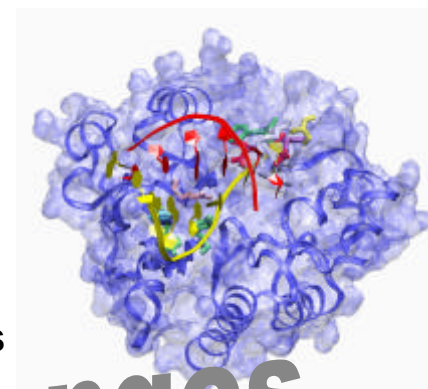
Sequence
Annotation

Promoter
T
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T
A
C
A
G
Q
T
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Y
C
C
C

Homology based
protein structure
prediction



Molecular
simulations

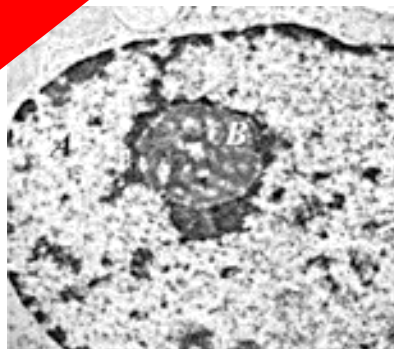


New Challenges

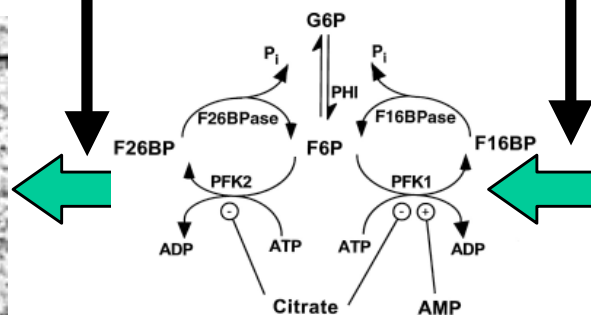
Data
integration

Pathway
simulations

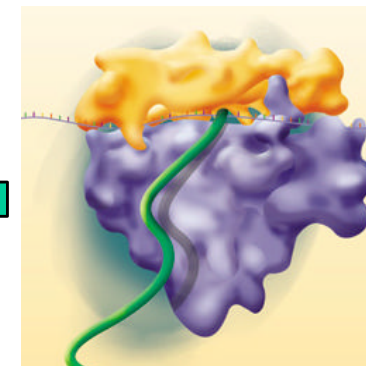
Network
analysis



Bacteria and cells



Metabolic pathways
and regulatory networks



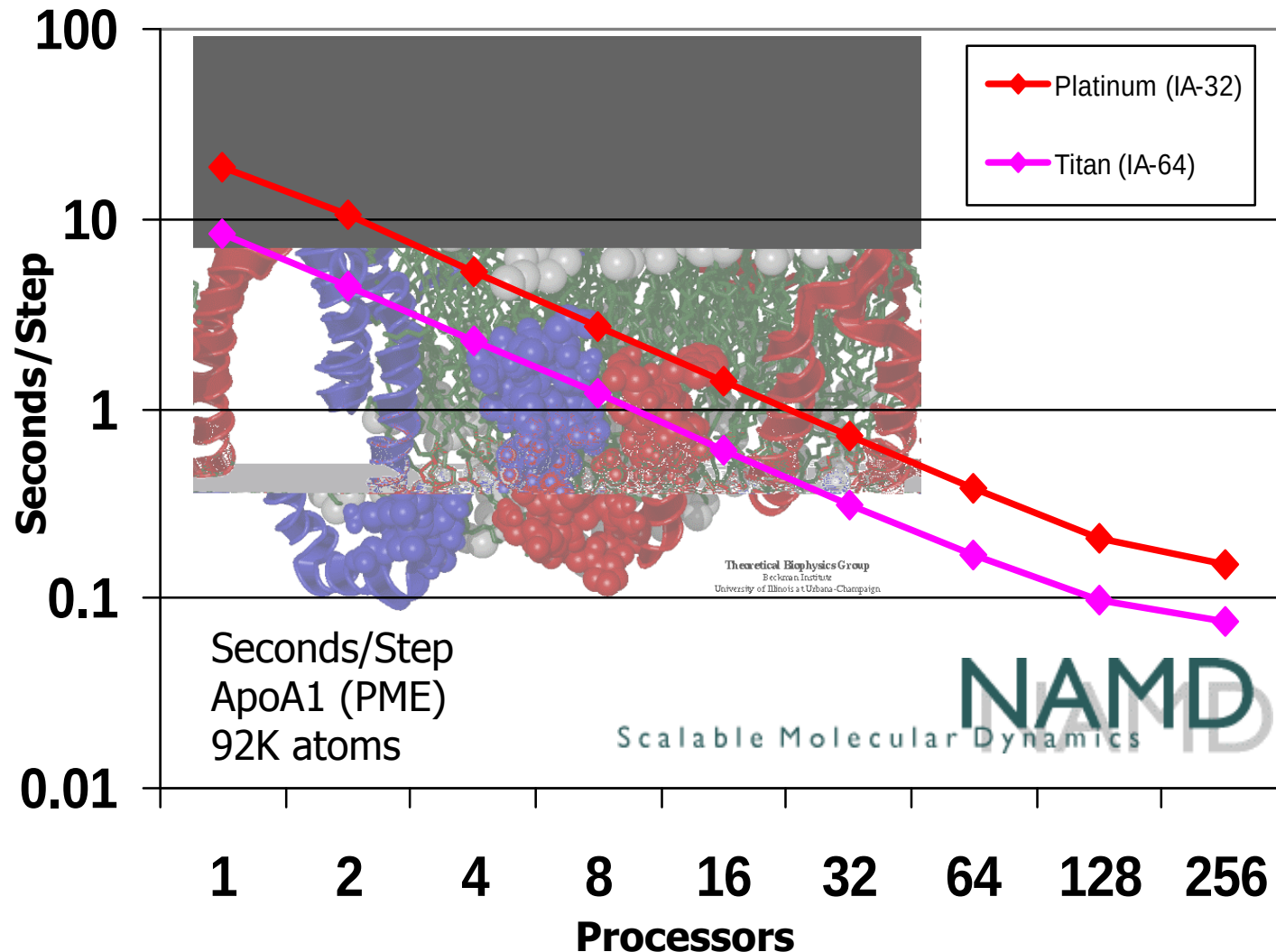
Multi-protein
machines

NCSA

National Center for Supercomputing Applications

National Computational Science ALLIANCE

NAMD: Scalable Molecular Dynamics



Simulation of large biomolecular systems on parallel computers

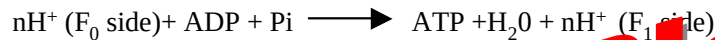
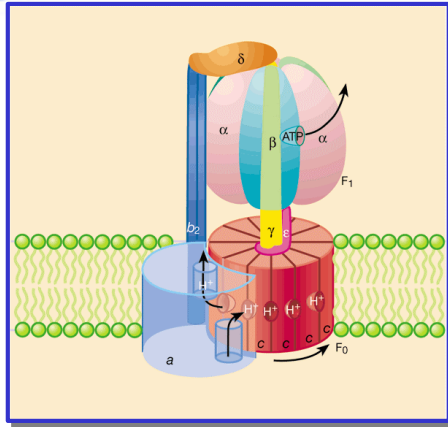
File compatible with CHARMM & AMBER

Message-driven and object-oriented design implemented with Charm++/Converse (from PPL at UIUC)

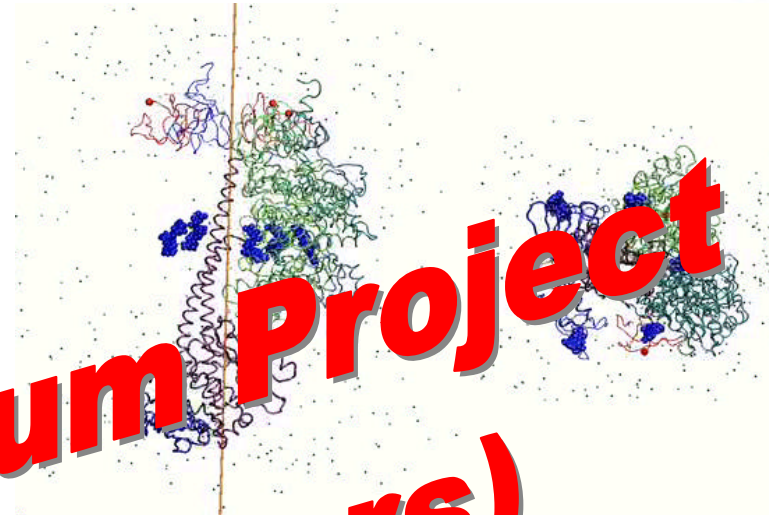
Ported to PACI systems, clusters, and desktop PCs

Available for **FREE**, includes source code

ATP Synthase F1 Stalk Rotation

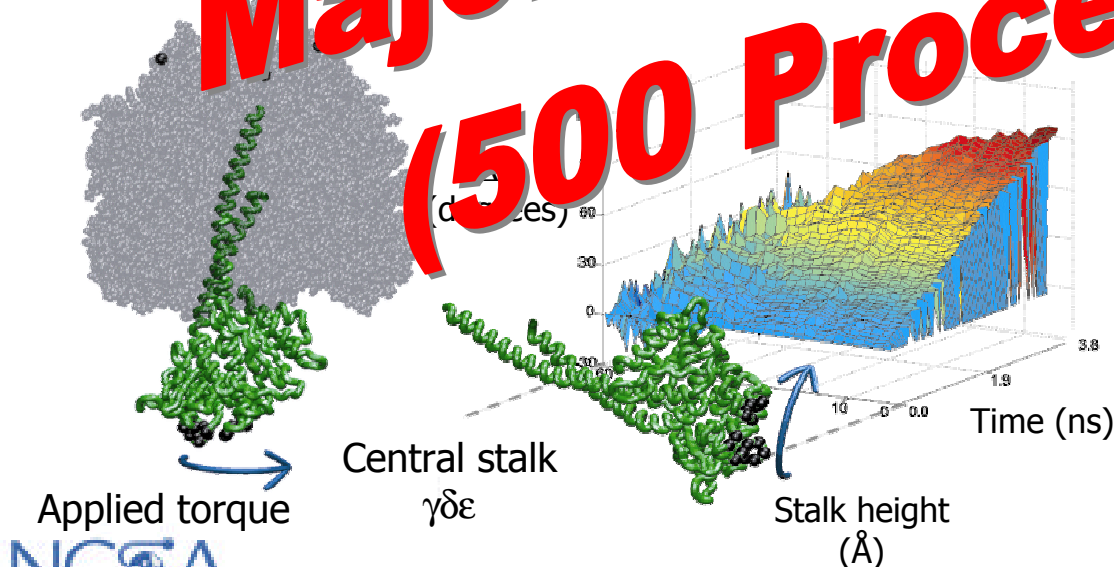


ATP synthase F₁ is a cellular machine that turns rotation into ATP synthesis



Trial simulation. Torque applied to 361 atoms at top of stalk; $w=10,800^\circ/\text{ns}$

Major Platinum Project
(500 Processors)

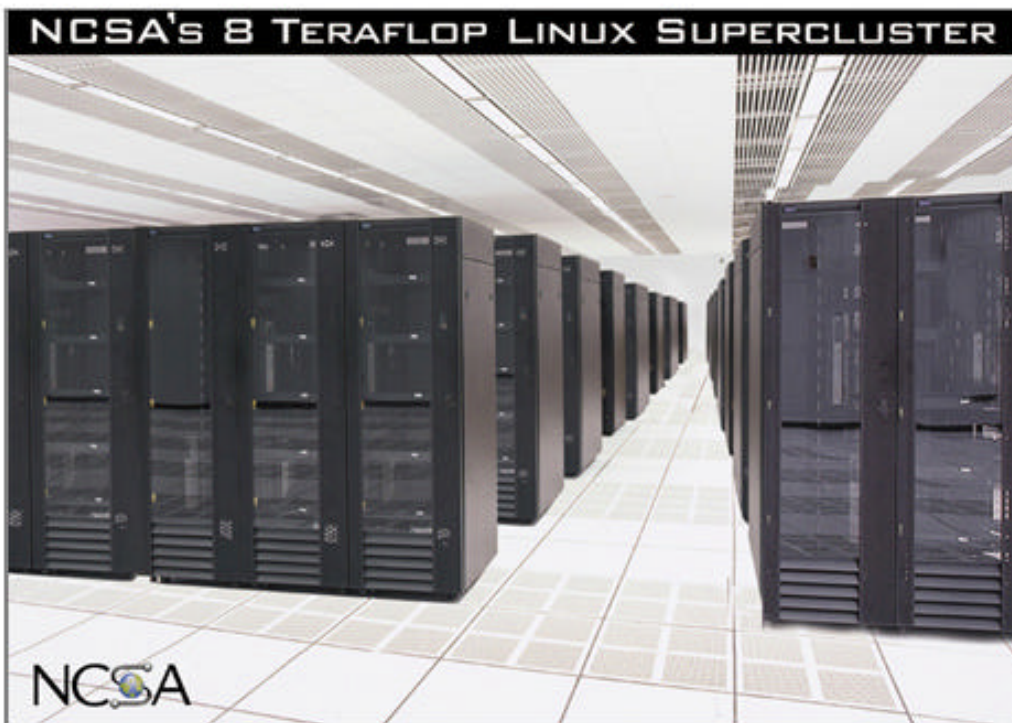


Production simulation. Torque applied to 16 atoms at bottom of stalk; $w=24^\circ/\text{ns}$. Rotation smoothly propagates along central stalk

The system contains 327,000 atoms. The 3.8 ns MD simulation (particle mesh Ewald electrostatics, periodic boundary conditions) required 135,000 SU on NCSA Platinum



TeraGrid: Linux Writ Large



- **TeraGrid partners**

- **NCSA**

- Dan Reed

- **SDSC**

- Fran Berman

- **Argonne**

- Rick Stevens

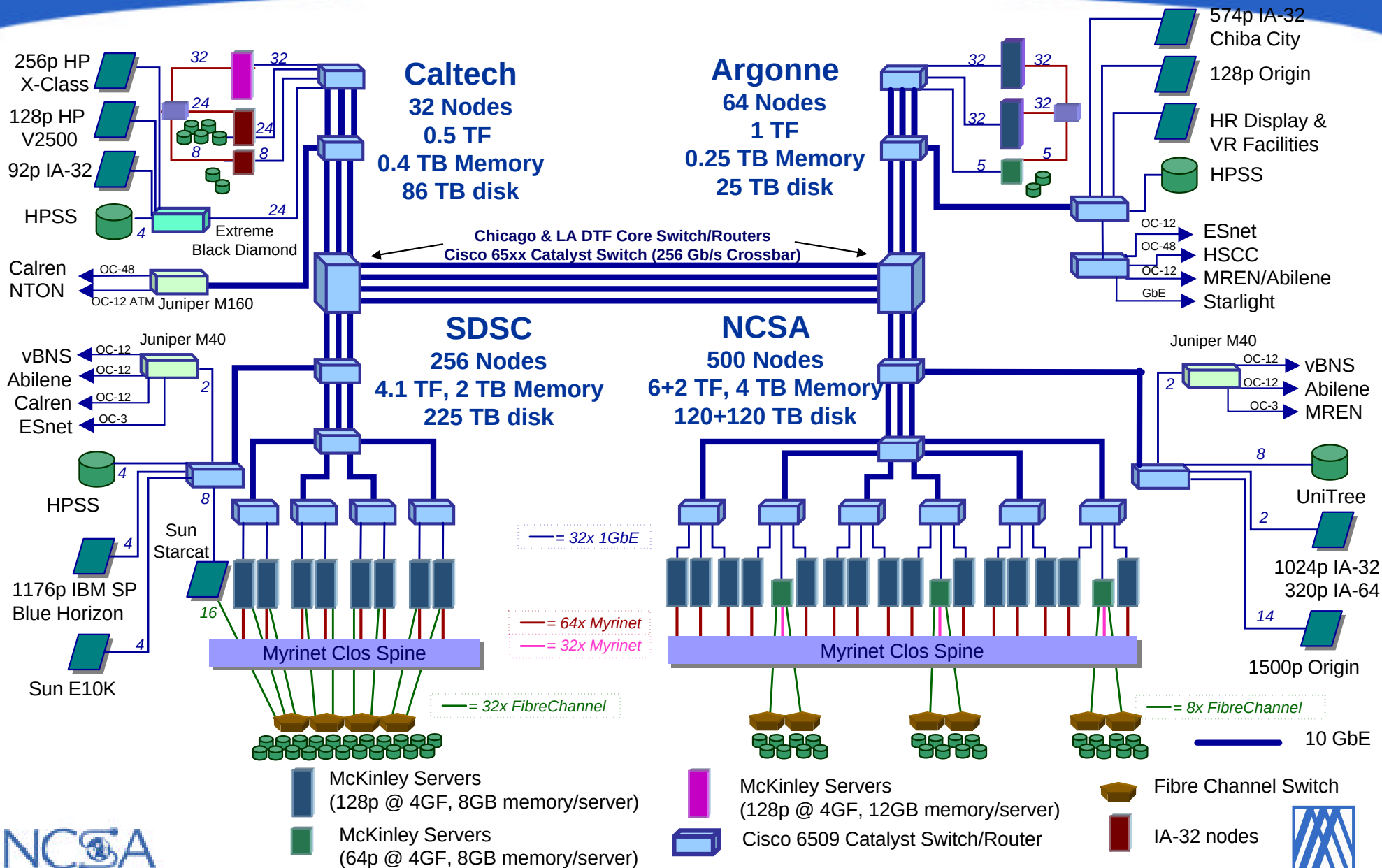
- Ian Foster

- **Caltech**

- Paul Messina

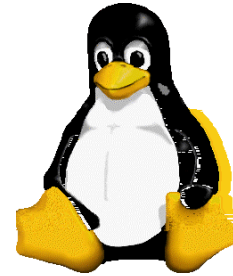


PACI Linux McKinley TeraGrid



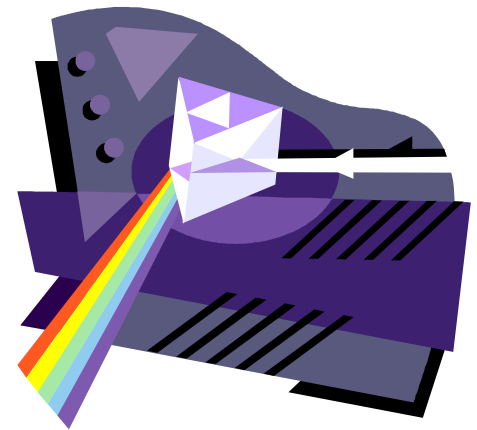
TeraGrid Strategic Partners

- **IBM**
 - cluster integration and RAS
 - GPFS parallel file system
- **Intel**
 - McKinley IA-64 software and compilers
- **Oracle/IBM**
 - data archive management and mining
- **Qwest**
 - 40 Gb/s DTF WAN backbone
- **Myricom**
 - cluster interconnect
- **SUN**
 - metadata service



TeraGrid Application Targets

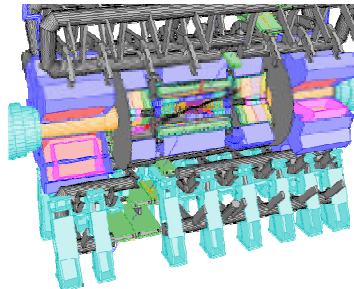
- **Multiple classes of user support**
 - each with differing implementation complexity
 - minimal change from current practice
 - new models, software, and applications
- **Usage exemplars**
 - “traditional” supercomputing made simpler
 - remote access to data archives and computers
 - distributed data archive access and correlation
 - remote rendering and visualization
 - remote sensor and instrument coupling



Challenges and Opportunities



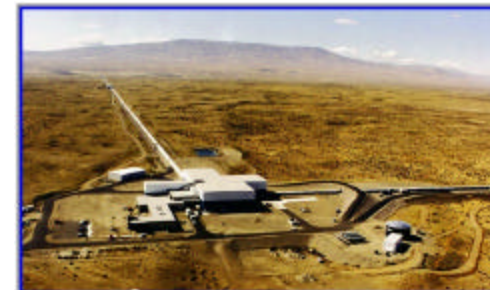
ALMA



ATLAS and CMS



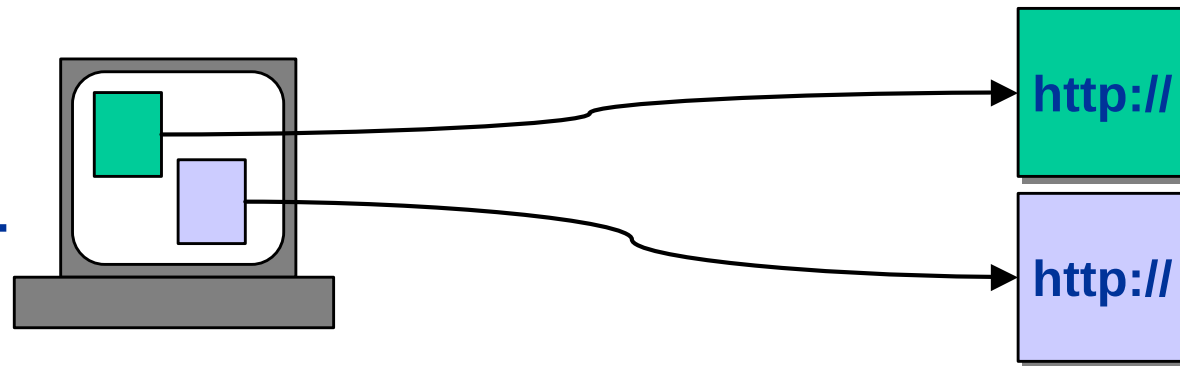
Sloan DSS



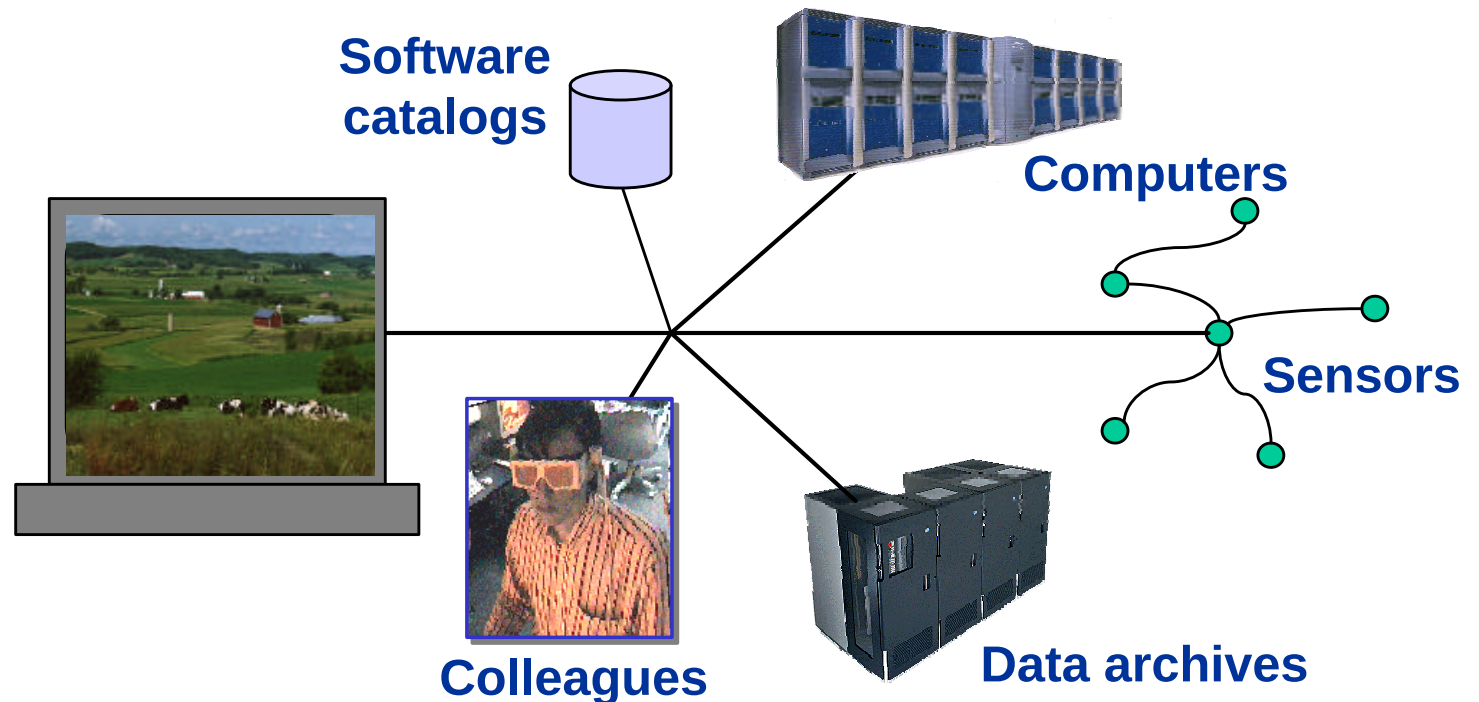
LIGO

Grids: The New Internet

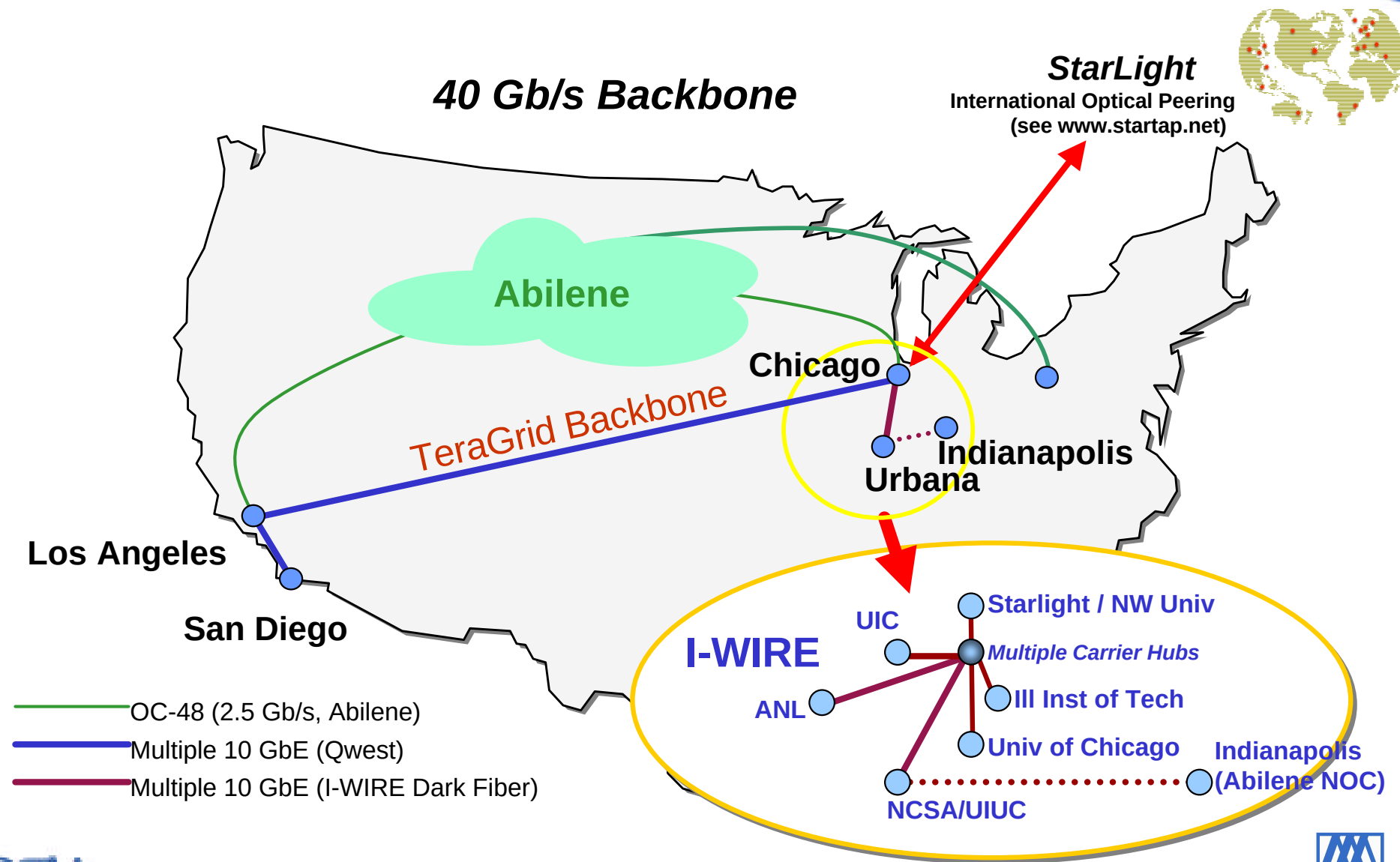
Web: Uniform access to HTML documents



Grid: Flexible, high-performance access to resources

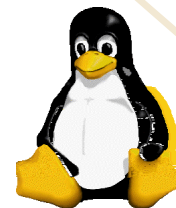
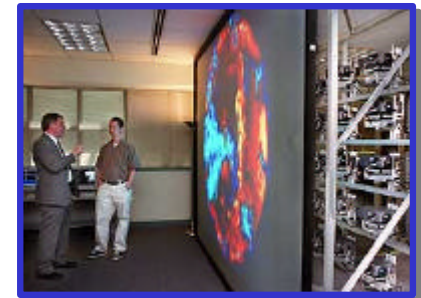
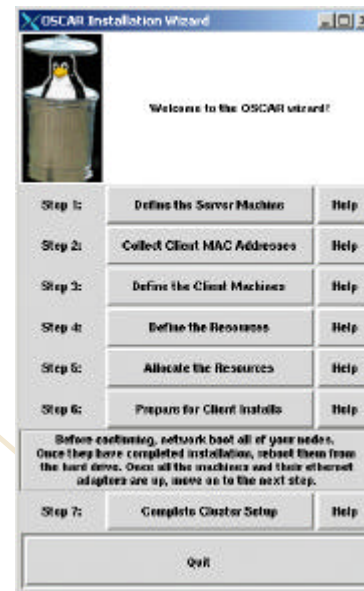


TeraGrid Network Backbone



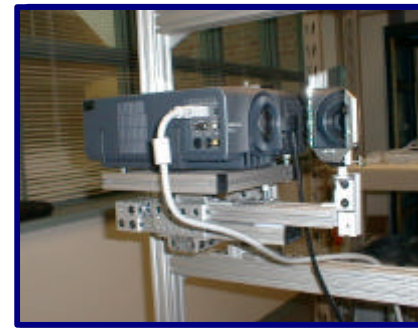
Alliance X-in-a-Box Initiatives

- **Access Grid**
 - Stevens/Childers (ANL)
- **Display Wall**
 - Baker (NCSA)
- **Clusters**
 - OSCAR/Pennington (NCSA)
 - <http://oscar.sourceforge.net>
 - <http://www.ncsa.uiuc.edu/TechFocus/Deployment/CiB/index.html>
- **Grid**
 - Butler (NCSA) and Foster (ANL)
- **Data**
 - Welge/Folk (NCSA)
- **Applications**
 - Crutcher (NCSA)



Scalable Display Walls

- **Large format with human scale**
 - commodity components
 - projectors, Linux cluster, cheap frames
 - high resolution (8192 x 3840 pixels)
 - now doubling in size



Petabytes and Beyond

- “Massive” data is being redefined

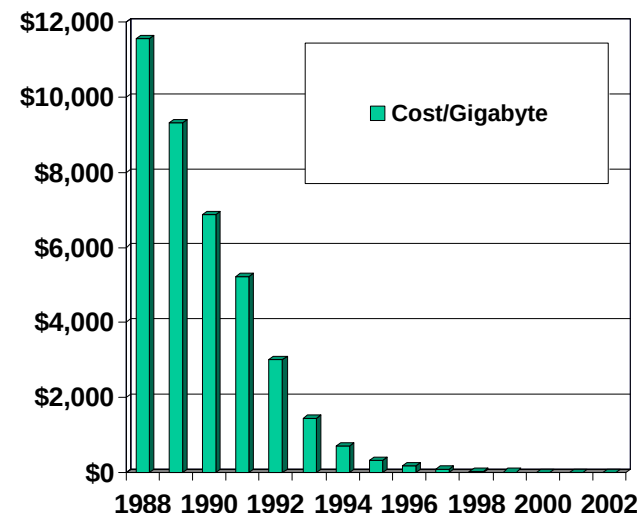
- O(\$5000) terabyte disk systems
 - commodity disks
 - software RAID
- petabyte disk farms very near
 - personal petabytes to follow

- Look at the science

- experimental (LHC, ALMA, LIGO, ...)
- computational (lots of great examples)
- petabytes/day coming RSN!

- Implications

- data explosion of unprecedented proportions
- data is *not* knowledge
- data mining is becoming critical



Petaops: Thinking Out of the Box

- **Riding the commodity technology curve**

- petaflop PC clusters (now)
 - \$1B-\$4B cost
 - 100 MW power
- new approaches
 - deep integration and packaging
 - new algorithms and software

- **Deskside teraflops**

- O(\$50K) teraflop systems
- empowering new research

- **The new commodities**

- IBM, Sony, Toshiba PlayStation3
 - 1000X PlayStation2 performance
 - \$400M investment
 - teraop processor design
- IBM research projects
 - Blue Gene and protein folding
- rethinking technology leverage



IBM Research

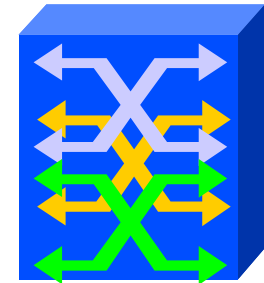


\$50,000 teraflops

The Ubiquitous Infosphere

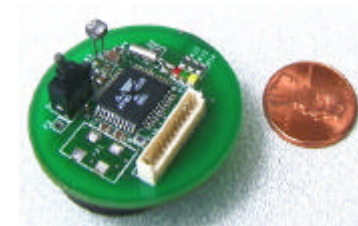
- **Terabit networking, the dominance of lambdas**

- Dense Wavelength Division Multiplexing (DWDM)
 - thousands of lambdas/fiber
- bandwidth growth to terabits/second
 - fiber being laid at 10,000 km/hour
- the “revenge” of circuit switching
 - wavelengths as instrument of exchange
- *the commodity cluster Grid writ large*



- **Mobile sensors, pixie dust gets real**

- micro-electro-mechanical systems (MEMS)
- wireless, sub-millimeter sensor/actuators
- *the commodity cluster Grid writ small*



From *In Vitro* to *In Silica*



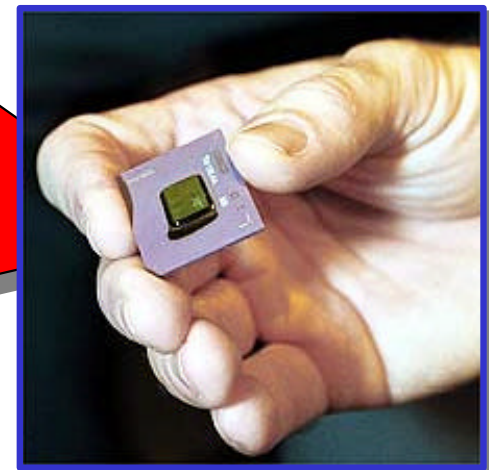
Associated Press

In Vitro



New York Times

In Vivo



Agence France-Presse

Teraflop in a Minute



NCSA IA-64 Cluster Deployment

